

ASIA INDEX STRATEGY

FTSE Taiwan Index Series Review and Flow Implications (June 2026)

What Happened? FTSE Russell announced the results of its quarterly index review for the FTSE TWSE Taiwan Index Series after the market closed on June 5 (Friday). All changes will take effect after market close on June 18 (Thursday).

Constituent Changes: For the **FTSE TWSE Taiwan 50**, **Zhen Ding (4958)**, **BizLink (3665)**, **Global Unichip (3443)**, and **Nan Ya PCB (8046)** will replace **China Steel (2002)**, **Formosa Plastics (1301)**, **Hotai Motor (2207)**, and **Caliway Biopharm (6919)**. For the **FTSE Taiwan Dividend+**, **Winbond Electronics (2344)**, **Nan Ya Plastics (1303)**, **Nanya Technology (2408)**, **Far Eastone (4904)**, and **Chung-Hsin Electric (1513)** will replace **Makalot Industrial (1477)**, **Tong Yang Industry (1319)**, **Tung Ho Steel (2006)**, and **Radiant Opto-Electronics (6176)**, with reweighting also causing notable flow impacts among existing constituents. Weight adjustments will be around 1.4% for Taiwan 50 and 23.6% for Taiwan Dividend+. (Exhibit 5)

Index Implications: Following the rebalancing, FTSE TWSE Taiwan 50 index's forward 12-month P/E and EPS growth (2026-27 CAGR) are expected to shift from 16.9x to 21.7x and 36.3% to 36.2%, while trailing dividend yield will remain unchanged (1.4%). For Taiwan Dividend+ index, the corresponding metrics would change from 16.2x to 13.4x (fP/E), 3.4% to 3.5% (D/Y), and 17.6% to 29.7% (EPS growth). (Exhibit 1)

Sector Implications: Banks (+US\$1.0bn), Chemiclas (+US\$760mn), Telcos (+US\$590mn), and Capital Goods (+US\$460mn) are poised to see the largest passive inflows, while Tech Hardware & Semis (-US\$2.4bn) may face the biggest funding outflows. These changes reflect rebalancing for existing and pending constituents in FTSE Taiwan 50/Dividend+, combined with potential flows from FTSE GEIS. Overall, we estimate that the **FTSE rebalancing could generate US\$13-14bn in gross two-way passive flows** for Taiwan. (Exhibit 2)

Historical vs. Current Patterns: FTSE Taiwan 50 additions have underperformed deletions ahead of the announcement, while Dividend+ additions have significantly outperformed; historical patterns suggest post-announcement outperformance for both, followed by a potential reversal after the changes become effective. (Exhibit 3)

Alvin So, CFA

+852-2978-1585 | alvin.so@gs.com
Goldman Sachs (Asia) L.L.C.

Timothy Moe, CFA

+65-6889-1199 | timothy.moe@gs.com
Goldman Sachs (Singapore) Pte

Kinger Lau, CFA

+852-2978-1224 | kinger.lau@gs.com
Goldman Sachs (Asia) L.L.C.

Sunil Koul

+44(20)7051-4931 | sunil.koul@gs.com
Goldman Sachs International

Exhibit 1: FTSE Russell announced the quarterly review results for the Taiwan Index Series, with 4/5 stock changes in the Taiwan 50/Dividend+, effective after market close on June 18

	FTSE Taiwan Index Series	Jun 2026 Review Summary	
		Taiwan 50	Dividend+
Stock Changes	Current # Constituents	50	49
	# Stocks Added / Deleted	4 / 4	5 / 4
	% Weight Rebalanced	1.4%	23.6%
Liquidity	Proforma Index Cap (US\$bn)	3,156	889
	Proforma % Change	+0.7%	+21.5%
	Proforma 3M ADVT (US\$bn)	15.9	8.3
	Proforma % Change	+9.5%	+35.2%
Valuations / Growth	NTM P/E (x)	16.9x /	16.2x /
	(Current / Proforma)	21.7x	13.4x
	LTM Dividend Yield (%)	1.4% /	3.4% /
	(Current / Proforma)	1.4%	3.5%
	2026-27 EPS Growth (%)	36.3% /	17.6% /
	(Current / Proforma)	36.2%	29.7%
Passive Flows (+FTSE GEIS)	Potential Gross Passive Buying + Selling (Two-way) (US\$mn)	2,637	10,862
Trading Pattern Prior to Announce.	Return Since 20D Prior to Ann. (Additions / Deletions)	-8.2% / +1.2%	+17.9% / +1.4%
	Share Turnover % Chg (Median 20D vs. Prior 3M) (Add / Del)	+5% / -15%	+52% / +40%

Note: Potential passive flows consider both FTSE Taiwan and Global Index rebalancing impact. Pricing is as of Jun 5, 2026

Source: FTSE Russell, FactSet, Refinitiv, Goldman Sachs Global Investment Research

Exhibit 2: Banks, Chemicals, Telcos, and Capital Goods are poised to see the largest passive inflows, while Tech Hardware & Semis may face the biggest funding outflows

Current vs. Proforma Sector Weights and Changes in FTSE TWSE Taiwan 50 / Dividend+, and Aggregate Potential Passive Flows

GICS Sector/Industry		Taiwan 50			Taiwan Dividend+			Passive Flows (US\$mn)		
		Current %	Proforma %	Chg (bps)	Current %	Proforma %	Chg (bps)	Gross Buying	Gross Selling	Net Flows
Financials	Banks	4.2%	4.2%	-1	16.8%	21.1%	+428	1,157	-168	989
	Insurance & Other Financials	3.1%	3.1%	+0	6.0%	5.8%	-17	164	-185	-21
	Real Estate	0.0%	0.0%	+0	0.0%	0.0%	+0	0	0	0
Domestic / Global Cyclical	Capital Goods	0.0%	0.4%	+39	0.3%	0.9%	+57	454	-1	453
	Transportation	0.3%	0.3%	-0	4.5%	4.7%	+28	65	-2	62
	Autos & Components	0.0%	0.0%	+0	1.1%	0.7%	-33	6	-87	-81
	Consumer Retail & Services	0.1%	0.0%	-14	0.8%	0.5%	-27	20	-193	-173
	Tech Hardware & Semis	90.0%	90.2%	+29	63.5%	53.1%	-1042	3,153	-5,518	-2,365
	Software & Services	0.0%	0.0%	+0	0.0%	0.0%	+0	0	0	0
Commodities	Internet/Media & Entertainment	0.0%	0.0%	+0	0.0%	0.0%	+0	0	0	0
	Energy	0.1%	0.1%	-0	0.0%	0.0%	+0	0	-5	-5
	Metals & Mining	0.2%	0.0%	-22	0.9%	0.7%	-25	11	-254	-243
Defensives	Chemicals & Other Materials	0.7%	0.5%	-21	0.7%	4.8%	+401	931	-177	754
	Consumer Staples	0.4%	0.4%	-0	2.0%	2.8%	+78	181	-1	180
	Health Care	0.1%	0.0%	-10	0.0%	0.0%	+0	2	-79	-77
	Utilities	0.0%	0.0%	+0	0.0%	0.0%	+0	0	0	0
	Telecommunication Services	0.9%	0.9%	-1	2.3%	4.8%	+255	592	-7	586

Note: Potential passive flows consider both FTSE Taiwan and Global Index rebalancing impact. Pricing is as of Jun 5, 2026

Source: FactSet, Refinitiv, EPFR, Goldman Sachs Global Investment Research

Exhibit 3: FTSE Taiwan 50: Additions have underperformed deletions in the two weeks ahead of the announcement; historical trends suggest post-announcement outperformance, followed by a potential reversal after the effective date

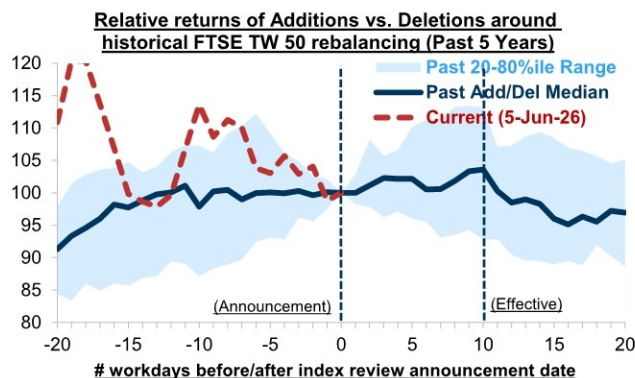
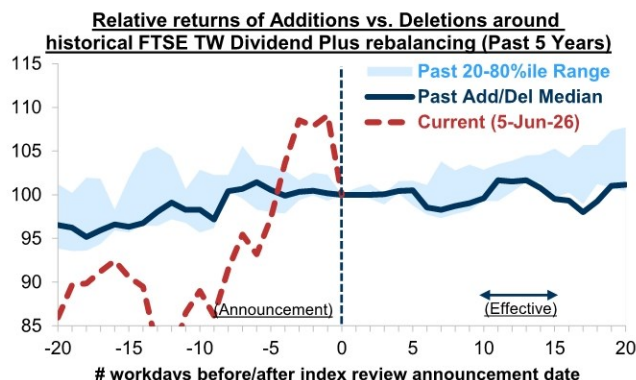


Exhibit 4: FTSE Taiwan Dividend+: Additions have significantly outperformed deletions; historical trends suggest moderate post-announcement outperformance to continue before the effective date



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 5: FTSE TWSE Taiwan: Stocks that may experience the most significant net passive buying or selling flows following the FTSE TWSE Taiwan and Global Equity Index Series (GEIS) rebalancing

BBG Ticker	Company Name	Sector	Listed Mkt Cap (US\$m)	Free Float Cap (US\$m)	3M ADVT (US\$m)	Potential Passive Flows (US\$m)			Net Passive Flows (US\$m)	Net Flows / 3M ADVT (days)	S3's Short Interest 200 Chg (pp)	Days to Cover (1M ADVT)	S3's HF Long Interest 200 Chg (pp)	Return since prior to ann.	Return since prior to ann.	Share Turnover % Chg (Median Prior 3M)	Corporate Event Dates	
						Taiwan 50	Taiwan Dividend +	All Cap / World									Expected Next Report	Ex-Dividend
FTSE Taiwan 50/Dividend+ Stocks with net passive buying following FTSE Taiwan and GEIS rebalancing																		
2344 TT	Winbond Electronics	Tech Hardware & Semis	25,672	17,489	803	(6)	953	(6)	941	1.2	(0.2%)	0.8	0.0%	51%	30%	52%	Aug 5	
1303 TT	Nan Ya Plastics	Chemicals & Other Materials	27,979	17,495	241	(3)	912	(1)	908	3.8	-	-	-	18%	21%	(32%)	Jul 10	Jul 8
2408 TT	Nanya Technology	Tech Hardware & Semis	43,314	16,334	1,114	3	833	6	842	0.8	(0.9%)	0.3	0.0%	31%	16%	52%	Jul 10	Jul 7
4904 TT	Far Eastone Telecom	Telecommunication Services	11,695	6,514	20	(1)	571	(0)	569	28.1	(0.0%)	0.5	0.0%	6%	6%	30%	Jul 10	Jul 8
4958 TT	Zhen Ding Technology	Tech Hardware & Semis	16,896	11,600	481	314	-	15	329	0.7	(1.9%)	0.6	0.0%	26%	(2%)	24%	Aug 12	Jun 8
3665 TT	BizLink	Capital Goods	13,361	11,366	234	321	-	0	322	1.4	(0.3%)	1.4	(0.0%)	(20%)	(1%)	(27%)	Aug 20	Jul 15
2887 TT	TS Financial	Banks	22,563	20,510	75	1	311	4	316	4.2	0.2%	2.4	0.1%	20%	20%	101%	Aug 31	
2884 TT	E.SUN Financial	Banks	17,195	15,207	41	(7)	304	(5)	292	7.2	0.1%	6.2	0.0%	3%	7%	34%	Aug 11	
3443 TT	Global Unichip	Tech Hardware & Semis	18,741	11,075	310	290	-	(11)	278	0.9	(0.2%)	0.3	0.0%	(15%)	(14%)	16%	Jul 31	Jun 8
2886 TT	Mega Financial	Banks	19,848	15,237	37	(3)	188	(0)	185	5.1	0.1%	1.6	0.0%	4%	5%	36%	Aug 31	
1216 TT	Uni-President Enterprises	Consumer Staples	13,291	10,995	31	(1)	181	1	180	5.7	0.2%	12.7	0.0%	1%	3%	57%	Aug 7	Jul 30
3231 TT	Wistron	Tech Hardware & Semis	17,790	16,426	247	(9)	190	(8)	173	0.7	0.4%	1.8	0.0%	17%	18%	72%	Aug 12	Jul 8
2890 TT	SinoPac Financial	Banks	15,269	13,256	31	2	150	4	157	5.0	(0.0%)	0.3	-	5%	12%	65%	Aug 24	
8046 TT	Nan Ya PCB	Tech Hardware & Semis	18,031	5,987	483	148	-	1	149	0.3	0.0%	0.3	0.0%	(1%)	(10%)	(5%)	Aug 6	Jun 25
2880 TT	Hua Nan Financial	Banks	16,675	11,489	32	(5)	157	(4)	148	4.6	0.7%	2.4	0.0%	5%	9%	106%	Aug 28	
1513 TT	Chung-Hsin Electric & Machinery	Capital Goods	2,990	2,417	46	-	131	(1)	130	2.8	3.5%	3.3	0.0%	8%	9%	81%	Aug 12	Jul 9
2883 TT	KGI Financial Holding	Insurance & Other Financials	14,782	13,823	46	(3)	133	(1)	129	2.8	0.1%	0.7	0.0%	21%	27%	50%	Aug 19	
2382 TT	Quanta Computer	Tech Hardware & Semis	49,596	34,197	323	(5)	97	(0)	92	0.3	3.3%	4.9	0.1%	15%	24%	141%	Aug 10	
2603 TT	Evergreen Marine (Taiwan)	Transportation	16,239	11,190	109	(2)	54	(0)	52	0.5	1.0%	7.2	-	10%	6%	171%	Aug 13	Jun 17
4938 TT	Pegatron	Tech Hardware & Semis	8,268	5,977	44	-	23	5	29	0.7	-	-	0.0%	15%	20%	28%	Aug 13	Jun 15
FTSE Taiwan 50/Dividend+ Stocks with net passive selling following FTSE Taiwan and GEIS rebalancing																		
2454 TT	MediaTek	Tech Hardware & Semis	225,822	203,496	1,107	(29)	(1,859)	(2)	(1,889)	(1.7)	(0.1%)	0.2	0.0%	18%	11%	(9%)	Jul 30	Jul 7
2303 TT	United Microelectronics	Tech Hardware & Semis	49,966	46,190	648	(8)	(1,310)	(1)	(1,319)	(2.0)	(0.5%)	1.8	-	44%	15%	176%	Jul 29	
3711 TT	ASE Technology	Tech Hardware & Semis	84,073	60,404	364	7	(898)	18	(873)	(2.4)	0.2%	2.3	0.0%	12%	3%	20%	Jul 31	
2330 TT	TSMC	Tech Hardware & Semis	1,965,697	1,828,486	2,515	(359)	-	(125)	(485)	(0.2)	0.0%	0.4	0.0%	3%	5%	(22%)	Jul 17	Jun 11
2449 TT	King Yuan Electronics	Tech Hardware & Semis	12,513	10,867	341	(2)	(295)	(1)	(298)	(0.9)	0.8%	1.5	0.0%	(0%)	5%	98%	Aug 10	
2301 TT	Lite-On Technology	Tech Hardware & Semis	18,292	15,218	208	(0)	(231)	2	(228)	(1.1)	(0.7%)	0.8	0.0%	13%	11%	96%	Jul 30	
2002 TT	China Steel	Metals & Mining	9,627	7,011	35	(178)	-	(4)	(181)	(5.3)	2.1%	11.2	0.0%	3%	6%	4%	Aug 10	
1301 TT	Formosa Plastics	Chemicals & Other Materials	10,278	7,157	48	(169)	-	(3)	(172)	(3.6)	0.6%	5.4	-	1%	8%	(34%)	Jul 10	Jun 30
2885 TT	Yuanta Financial Holdings	Insurance & Other Financials	27,074	21,814	47	3	(173)	6	(164)	(3.5)	0.1%	1.7	(0.0%)	16%	19%	19%	Jul 10	
2891 TT	CTBC Financial	Banks	42,528	37,145	92	(5)	(136)	0	(141)	(1.5)	0.0%	4.3	0.0%	18%	16%	35%	Aug 25	
2207 TT	Hotai Motor	Consumer Retail & Services	8,596	4,349	8	(114)	-	3	(111)	(13.8)	0.2%	25.1	0.1%	2%	7%	19%	Aug 12	Jun 12
1477 TT	Makalot Industrial	Consumer Retail & Services	1,803	1,581	17	-	(79)	0	(79)	(4.6)	2.1%	6.3	0.0%	3%	4%	64%	Jul 31	
1319 TT	Tong Yang Industry	Autos & Components	1,814	1,099	10	-	(81)	2	(79)	(7.8)	1.9%	8.0	0.0%	22%	20%	16%	Aug 7	Jun 22
6919 TT	Calway Biopharms	Health Care	4,771	3,033	34	(79)	-	2	(77)	(2.3)	-	-	0.0%	(8%)	(8%)	(43%)	Aug 10	
2006 TT	Tung Ho Steel	Metals & Mining	1,564	962	7	-	(70)	(0)	(70)	(10.5)	1.1%	5.2	-	(1%)	(2%)	(31%)	Jul 29	
6176 TT	Radiant Opto-Electronics	Tech Hardware & Semis	1,500	1,421	24	-	(64)	(0)	(65)	(2.7)	0.8%	3.8	0.0%	(0%)	(3%)	95%	Jul 31	Jun 1
2357 TT	Asustek Computer	Tech Hardware & Semis	21,057	18,876	107	(9)	(10)	(8)	(28)	(0.3)	(0.7%)	9.1	0.0%	38%	32%	48%	Aug 13	
Taiwan 50 Median			21,621	16,380	135						0.1%	1.5	0.0%	6%	7%	34%		
Taiwan Dividend+ Median			8,553	7,026	47						0.1%	2.7	0.0%	11%	7%	64%		

Note: Free Float Cap is calculated based on the average of estimates from Bloomberg, FactSet and Refinitiv. Passive funds AUM is derived from EPFR and FactSet ownership data. Pricing is as of Jun 5, 2026

Names highlighted in blue or grey indicate additions or deletions

Source: FactSet, Refinitiv, Bloomberg, EPFR, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Alvin So, CFA, Timothy Moe, CFA, Kinger Lau, CFA and Sunil Koul, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Alvin So, CFA Goldman Sachs (Asia) L.L.C., Timothy Moe, CFA Goldman Sachs (Singapore) Pte, Kinger Lau, CFA Goldman Sachs (Asia) L.L.C., Sunil Koul Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is

defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your

individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.